LAB EXPERIEMENT RECORD

On

UCS4001

DATABASE MANAGEMENT SYSTEM

SCHOOL OF COMPUTER SCIENCE AND ENGINERING

IILM UNIVERSITY

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EXPERIMENT NO. 01

Perform following Query :

1. Create Employee_master table with data using Employee table.

```
MariaDB [ritika0748]> CREATE TABLE EMPLOYEE_MASTER AS SELECT * FROM EMPLOYEE;
Query OK, 14 rows affected (0.034 sec)
Records: 14  Duplicates: 0  Warnings: 0

MariaDB [ritika0748]> select * from employee_master;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
14 rows in set (0.001 sec)
```

2. Delete all record into Employee_master whose DeptNo is 10.

```
MariaDB [ritika0748]>  DELETE FROM employee_master where DEPTNO = 10;
Query OK, 1 row affected (0.019 sec)

MariaDB [ritika0748]> select * from employee_master;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.001 sec)
```

3. Update 10% in his salary of DEPTNO 20 into Employee_Master.

```
MariaDB [ritika0748]> UPDATE employee_master
    -> SET sal = sal * 1.1
    -> WHERE DEPTNO = 20;
Query OK, 6 rows affected (0.003 sec)
Rows matched: 6  Changed: 6  Warnings: 0

MariaDB [ritika0748]> select * from employee_master;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300 | NULL |     20 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.001 sec)
```

4. Alter SAL with size 10,2 in Employee_Master.

```
MariaDB [ritika0748]> ALTER TABLE employee_master
    -> MODIFY COLUMN sal DECIMAL(10,2);
Query OK, 13 rows affected (0.053 sec)
Records: 13  Duplicates: 0  Warnings: 0

MariaDB [ritika0748]> select * from employee_master;
+-------+--------+-----------+------+------------+---------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | sal     | COMM | DEPTNO |
+-------+--------+-----------+------+------------+---------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  880.00 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600.00 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250.00 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3273.00 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250.00 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 2850.00 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2695.00 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000.00 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5500.00 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500.00 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1210.00 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950.00 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3300.00 | NULL |     20 |
+-------+--------+-----------+------+------------+---------+------+--------+
13 rows in set (0.003 sec)
```

5. Drop Employee_master Table.

```
MariaDB [ritika0748]> DROP TABLE employee_master;
Query OK, 0 rows affected (0.011 sec)

MariaDB [ritika0748]> SELECT * from employee_master;
ERROR 1146 (42S02): Table 'ritika0748.employee_master' doesn't exist
MariaDB [ritika0748]> show tables;
+----------------------+
| Tables_in_ritika0748 |
+----------------------+
| department           |
| employee             |
+----------------------+
```

EXPERIMENT NO. 02

Perform Following Query using Employee Table. (Retrieving Data) :

1. List all distinct job in Employee.

```
MariaDB [ritika0748]> SELECT DISTINCT JOB from employee;
+-----------+
| JOB       |
+-----------+
| CLERK     |
| SALESMAN  |
| MANAGER   |
| ANALYST   |
| PRESIDENT |
+-----------+
5 rows in set (0.001 sec)
```

2. List all information about employee in Department Number 30.

```
MariaDB [ritika0748]> SELECT * from employee
    -> where deptNo = 30;
+-------+--------+----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+----------+------+------------+------+------+--------+
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 |  950 | NULL |     30 |
+-------+--------+----------+------+------------+------+------+--------+
6 rows in set (0.001 sec)
```

3. Find all department number with department names greater than 20.

```
MariaDB [ritika0748]> SELECT * from department
    -> where deptNo > 20;
+--------+------------+
| DeptNo | dName      |
+--------+------------+
|     30 | Sales      |
|     40 | Operations |
+--------+------------+
2 rows in set (0.001 sec)
```

4. Find all information about all the managers as well as the clerks in department 30.

```
MariaDB [ritika0748]> SELECT * from employee
    -> WHERE JOB in ("manager" , "clerk") and deptNo = 30;
+-------+-------+---------+------+------------+------+------+--------+
| EMPNO | ENAME | JOB     | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+-------+---------+------+------------+------+------+--------+
|  7698 | BLAKE | MANAGER | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7900 | JAMES | CLERK   | 7698 | 1981-12-03 |  950 | NULL |     30 |
+-------+-------+---------+------+------------+------+------+--------+
2 rows in set (0.001 sec)
```

5. List the Employee name, Employee numbers and department of all clerks.

```
MariaDB [ritika0748]> select empNo, eName, deptNo from employee
    -> where job = "clerk";
+-------+--------+--------+
| empNo | eName  | deptNo |
+-------+--------+--------+
|  7369 | SMITH  |     20 |
|  7876 | ADAMS  |     20 |
|  7900 | JAMES  |     30 |
|  7934 | MILLER |     10 |
+-------+--------+--------+
4 rows in set (0.001 sec)
```

6. Find all managers not in department 30.

```
MariaDB [ritika0748]> select empNo, eName from employee
    -> where job = "manager" and deptNo != 30;
+-------+-------+
| empNo | eName |
+-------+-------+
|  7566 | JONES |
|  7782 | CLARK |
+-------+-------+
2 rows in set (0.002 sec)
```

7. List information about all Employees in department 10 who are not manager or clerks.

```
MariaDB [ritika0748]> select * from employee
    -> where job not in ("Manager", "Clerk") and deptNO = 10;
Empty set (0.001 sec)
```

8. Find Employees and jobs earning between 1200 and 1400.

```
MariaDB [ritika0748]> select ename, job from employee
    -> where sal >= 1200 and sal <= 1400;
+--------+----------+
| ename  | job      |
+--------+----------+
| WARD   | SALESMAN |
| MARTIN | SALESMAN |
| MILLER | CLERK    |
+--------+----------+
3 rows in set (0.001 sec)
```

9. List Name and Department Number of employee who are clerks, analyst or salesman.

```
MariaDB [ritika0748]> select ename, job, deptNo from employee
    -> where job in ("clerk", "analyst", "salesman");
+--------+----------+--------+
| ename  | job      | deptNo |
+--------+----------+--------+
| SMITH  | CLERK    |     20 |
| ALLEN  | SALESMAN |     30 |
| WARD   | SALESMAN |     30 |
| MARTIN | SALESMAN |     30 |
| SCOTT  | ANALYST  |     40 |
| TURNER | SALESMAN |     30 |
| ADAMS  | CLERK    |     20 |
| JAMES  | CLERK    |     30 |
| FORD   | ANALYST  |     20 |
| MILLER | CLERK    |     10 |
+--------+----------+--------+
10 rows in set (0.001 sec)
```

10. List Name and Department Number of employee whose names began with M.

```
MariaDB [ritika0748]> select * from employee
    -> where ename LIKE "m%";
+-------+--------+----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+----------+------+------------+------+------+--------+
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7934 | MILLER | CLERK    | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+----------+------+------------+------+------+--------+
2 rows in set (0.001 sec)
```

EXPERIMENT NO. 03

1. List all employees and jobs in Department 30 in descending order by salary.

```
MariaDB [ritika0748]> select * from employee
    -> where DEPTNO = 30
    -> Order by SAL desc;
+-------+--------+----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+----------+------+------------+------+------+--------+
|  7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 |  950 | NULL |     30 |
+-------+--------+----------+------+------------+------+------+--------+
6 rows in set (0.008 sec)
```

2. List job and Department Number of employees whose name are five letters long begin with “A” and end with “N”.

```
MariaDB [ritika0748]> select ENAME, JOB, DEPTNO from employee
    -> where ENAME LIKE "A___N";
+-------+----------+--------+
| ENAME | JOB      | DEPTNO |
+-------+----------+--------+
| ALLEN | SALESMAN |     30 |
+-------+----------+--------+
1 row in set (0.001 sec)
```

3. Display the name of employees whose name start with alphabet S.

```
MariaDB [ritika0748]> select ENAME, JOB, DEPTNO from employee
    -> where ENAME LIKE "A___N";
+-------+----------+--------+
| ENAME | JOB      | DEPTNO |
+-------+----------+--------+
| ALLEN | SALESMAN |     30 |
+-------+----------+--------+
1 row in set (0.001 sec)
```

4. Display the names of employees whose name ends with alphabet S.

```
MariaDB [ritika0748]> select ENAME, JOB, DEPTNO from employee
    -> where ename LIKE "%S";
+-------+---------+--------+
| ENAME | JOB     | DEPTNO |
+-------+---------+--------+
| JONES | MANAGER |     20 |
| ADAMS | CLERK   |     20 |
| JAMES | CLERK   |     30 |
+-------+---------+--------+
3 rows in set (0.001 sec)
```

5. Display the names of employees working in department number 10 or 20 or 40 or employees working as clerks, salesman or analyst.

```
MariaDB [ritika0748]>  select * from employee
    -> where DEPTNO IN (10,20,40)
    -> OR JOB IN ("Clerk", "Salesman", "Analyst");
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1500 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 |  950 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3000 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
13 rows in set (0.001 sec)
```

6. Display employee number and names for employees who earn commission.

```
MariaDB [ritika0748]>  select ENAME, EMPNO, COMM from employee
    -> where COMM IS not null AND COMM > 0;
+--------+-------+------+
| ENAME  | EMPNO | COMM |
+--------+-------+------+
| ALLEN  |  7499 |  300 |
| WARD   |  7521 |  300 |
| MARTIN |  7654 | 1400 |
+--------+-------+------+
3 rows in set (0.001 sec)
```

7. Display employee number and total salary for each employee. 1,2

```
MariaDB [ritika0748]>  select ENAME, EMPNO, SAL + CASE
    -> WHEN comm is NULL then 0
    -> ELSE comm
    -> END  AS TOTALSAL from employee;
+--------+-------+----------+
| ENAME  | EMPNO | TOTALSAL |
+--------+-------+----------+
| SMITH  |  7369 |      800 |
| ALLEN  |  7499 |     1900 |
| WARD   |  7521 |     1550 |
| JONES  |  7566 |     2975 |
| MARTIN |  7654 |     2650 |
| BLAKE  |  7698 |     2850 |
| CLARK  |  7782 |     2450 |
| SCOTT  |  7788 |     3000 |
| KING   |  7839 |     5000 |
| TURNER |  7844 |     1500 |
| ADAMS  |  7876 |     1100 |
| JAMES  |  7900 |      950 |
| FORD   |  7902 |     3000 |
| MILLER |  7934 |     1300 |
+--------+-------+----------+
14 rows in set (0.001 sec)
```

```
MariaDB [ritika0748]>  select ENAME, EMPNO, SAL + IFNULL(comm, 0) AS TotalSal from employee;
+--------+-------+----------+
| ENAME  | EMPNO | TotalSal |
+--------+-------+----------+
| SMITH  |  7369 |      800 |
| ALLEN  |  7499 |     1900 |
| WARD   |  7521 |     1550 |
| JONES  |  7566 |     2975 |
| MARTIN |  7654 |     2650 |
| BLAKE  |  7698 |     2850 |
| CLARK  |  7782 |     2450 |
| SCOTT  |  7788 |     3000 |
| KING   |  7839 |     5000 |
| TURNER |  7844 |     1500 |
| ADAMS  |  7876 |     1100 |
| JAMES  |  7900 |      950 |
| FORD   |  7902 |     3000 |
| MILLER |  7934 |     1300 |
+--------+-------+----------+
14 rows in set (0.002 sec)
```

8. Display employee number and annual salary for each employee.

```
MariaDB [ritika0748]>  select ENAME, EMPNO, (SAL + IFNULL(comm, 0)) * 12
    -> AS AnnualSal from employee;
+--------+-------+-----------+
| ENAME  | EMPNO | AnnualSal |
+--------+-------+-----------+
| SMITH  |  7369 |      9600 |
| ALLEN  |  7499 |     22800 |
| WARD   |  7521 |     18600 |
| JONES  |  7566 |     35700 |
| MARTIN |  7654 |     31800 |
| BLAKE  |  7698 |     34200 |
| CLARK  |  7782 |     29400 |
| SCOTT  |  7788 |     36000 |
| KING   |  7839 |     60000 |
| TURNER |  7844 |     18000 |
| ADAMS  |  7876 |     13200 |
| JAMES  |  7900 |     11400 |
| FORD   |  7902 |     36000 |
| MILLER |  7934 |     15600 |
+--------+-------+-----------+
14 rows in set (0.001 sec)
```

```
MariaDB [ritika0748]>  select ENAME, EMPNO, (SAL + IFNULL(comm, 0)) * 12
    -> AS AnnualSal from employee;
+--------+-------+-----------+
| ENAME  | EMPNO | AnnualSal |
+--------+-------+-----------+
| SMITH  |  7369 |      9600 |
| ALLEN  |  7499 |     22800 |
| WARD   |  7521 |     18600 |
| JONES  |  7566 |     35700 |
| MARTIN |  7654 |     31800 |
| BLAKE  |  7698 |     34200 |
| CLARK  |  7782 |     29400 |
| SCOTT  |  7788 |     36000 |
| KING   |  7839 |     60000 |
| TURNER |  7844 |     18000 |
| ADAMS  |  7876 |     13200 |
| JAMES  |  7900 |     11400 |
| FORD   |  7902 |     36000 |
| MILLER |  7934 |     15600 |
+--------+-------+-----------+
14 rows in set (0.001 sec)
```

9. Display the names of all employees working as clerks and drawing a salary more than 3,000.

```
MariaDB [ritika0748]>  select ENAME, SAL from employee
    -> where job = "Clerk" and sal > 3000;
Empty set (0.001 sec)
```

---

10. Display the names of employees who are working as clerk, salesman or analyst and drawing a salary more than 3,000.

```
MariaDB [ritika0748]>  select ENAME, JOB from employee
    -> where JOB IN ("Clerk", "Salesman", "Analyst")
    -> AND SAL >= 3000 ;
+-------+---------+
| ENAME | JOB     |
+-------+---------+
| SCOTT | ANALYST |
| FORD  | ANALYST |
+-------+---------+
2 rows in set (0.001 sec)
```

EXPERIMENT NO. 04

1. Display the list of employees who have joined the company before 30th June 80 or after 31st Dec 81.

```
MariaDB [ritika0748]> select * from employee
    -> where hiredate < '1980-06-30' OR hiredate > '1981-12-31';
+-------+--------+---------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB     | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+---------+------+------------+------+------+--------+
|  7788 | SCOTT  | ANALYST | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7876 | ADAMS  | CLERK   | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7934 | MILLER | CLERK   | 7782 | 1982-01-23 | 1300 | NULL |     10 |
+-------+--------+---------+------+------------+------+------+--------+
3 rows in set (0.001 sec)
```

2. Display the names of employees whose names have second alphabet A in their names.

```
MariaDB [ritika0748]> select ename from employee
    -> where ename like "_A%";
+--------+
| ename  |
+--------+
| WARD   |
| MARTIN |
| JAMES  |
+--------+
3 rows in set (0.005 sec)
```

3. Display the names of employees whose name is exactly five characters in length

```
MariaDB [ritika0748]> select * from employee
    -> where length(ENAME) = 5;
+-------+-------+----------+------+------------+------+------+--------+
| EMPNO | ENAME | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+-------+----------+------+------------+------+------+--------+
|  7369 | SMITH | CLERK    | 7902 | 1980-12-17 |  800 | NULL |     20 |
|  7499 | ALLEN | SALESMAN | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7566 | JONES | MANAGER  | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7698 | BLAKE | MANAGER  | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK | MANAGER  | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7788 | SCOTT | ANALYST  | 7566 | 1982-12-09 | 3000 | NULL |     40 |
|  7876 | ADAMS | CLERK    | 7788 | 1983-01-12 | 1100 | NULL |     20 |
|  7900 | JAMES | CLERK    | 7698 | 1981-12-03 |  950 | NULL |     30 |
+-------+-------+----------+------+------------+------+------+--------+
8 rows in set (0.001 sec)
```

```
MariaDB [ritika0748]> select ename from employee
    -> where ename like "_____";
+-------+
| ename |
+-------+
| SMITH |
| ALLEN |
| JONES |
| BLAKE |
| CLARK |
| SCOTT |
| ADAMS |
| JAMES |
+-------+
8 rows in set (0.001 sec)
```

4. Display the names of employees whose names have second alphabet A in their names.

```
MariaDB [ritika0748]> select * from employee
    -> where ename LIKE '_A%';
+-------+--------+----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB      | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+----------+------+------------+------+------+--------+
|  7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7900 | JAMES  | CLERK    | 7698 | 1981-12-03 |  950 | NULL |     30 |
+-------+--------+----------+------+------------+------+------+--------+
3 rows in set (0.002 sec)
```

---

5. Display the names of employees who are not working as salesman or clerk or analyst.

```
MariaDB [ritika0748]> select * from employee
    -> where JOB NOT IN ("clerk", "salesman", "analyst") ;
+-------+-------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+-------+-----------+------+------------+------+------+--------+
|  7566 | JONES | MANAGER   | 7839 | 1981-04-02 | 2975 | NULL |     20 |
|  7698 | BLAKE | MANAGER   | 7839 | 1981-05-01 | 2850 | NULL |     30 |
|  7782 | CLARK | MANAGER   | 7839 | 1981-06-09 | 2450 | NULL |     20 |
|  7839 | KING  | PRESIDENT | NULL | 1981-11-17 | 5000 | NULL |     20 |
+-------+-------+-----------+------+------------+------+------+--------+
4 rows in set (0.001 sec)
```

6. Display the name of the employee along with their annual salary (sal*12). The name of the employee earning highest salary should appear first.

```
MariaDB [ritika0748]> select eName, Sal*12 AS AnnualSalary from employee
    -> Order by AnnualSalary desc;
+--------+--------------+
| eName  | AnnualSalary |
+--------+--------------+
| KING   |        60000 |
| SCOTT  |        36000 |
| FORD   |        36000 |
| JONES  |        35700 |
| BLAKE  |        34200 |
| CLARK  |        29400 |
| ALLEN  |        19200 |
| TURNER |        18000 |
| MILLER |        15600 |
| MARTIN |        15000 |
| WARD   |        15000 |
| ADAMS  |        13200 |
| JAMES  |        11400 |
| SMITH  |         9600 |
+--------+--------------+
14 rows in set (0.001 sec)
```

7. Display name, sal, hra, pf, da, totalsal for each employee. The output should be in the order of total sal, hra 15% of sal, da 10% of sal, pf 5% of sal. Total salary will be (sal*hra*da)-pf.

```
MariaDB [ritika0748]> select SAL * (SAL * 0.15) * (SAL * 0.1) - (SAL * 0.05) AS TotalSal,
    -> SAL * 0.15 AS HRA,
    -> SAL * 0.1 AS DA,
    -> SAL * 0.05 AS PF
    -> FROM employee;
+---------------+--------+-------+--------+
| TotalSal      | HRA    | DA    | PF     |
+---------------+--------+-------+--------+
|    7679960.000 | 120.00 |  80.0 |  40.00 |
|   61439920.000 | 240.00 | 160.0 |  80.00 |
|   29296812.500 | 187.50 | 125.0 |  62.50 |
|  394958991.875 | 446.25 | 297.5 | 148.75 |
|   29296812.500 | 187.50 | 125.0 |  62.50 |
|  347236732.500 | 427.50 | 285.0 | 142.50 |
|  220591752.500 | 367.50 | 245.0 | 122.50 |
|  404999850.000 | 450.00 | 300.0 | 150.00 |
| 1874999750.000 | 750.00 | 500.0 | 250.00 |
|   50624925.000 | 225.00 | 150.0 |  75.00 |
|   19964945.000 | 165.00 | 110.0 |  55.00 |
|   12860577.500 | 142.50 |  95.0 |  47.50 |
|  404999850.000 | 450.00 | 300.0 | 150.00 |
|   32954935.000 | 195.00 | 130.0 |  65.00 |
+---------------+--------+-------+--------+
14 rows in set (0.002 sec)
```

```
MariaDB [ritika0748]> select ENAME, SAL * (SAL * 0.15) * (SAL * 0.1) - (SAL * 0.05) AS TotalSal,
    -> SAL * 0.15 AS HRA,
    -> SAL * 0.1 AS DA,
    -> SAL * 0.05 AS PF
    -> FROM employee
    -> Order by TotalSal desc;
+--------+---------------+--------+-------+--------+
| ENAME  | TotalSal      | HRA    | DA    | PF     |
+--------+---------------+--------+-------+--------+
| KING   | 1874999750.000 | 750.00 | 500.0 | 250.00 |
| SCOTT  |  404999850.000 | 450.00 | 300.0 | 150.00 |
| FORD   |  404999850.000 | 450.00 | 300.0 | 150.00 |
| JONES  |  394958991.875 | 446.25 | 297.5 | 148.75 |
| BLAKE  |  347236732.500 | 427.50 | 285.0 | 142.50 |
| CLARK  |  220591752.500 | 367.50 | 245.0 | 122.50 |
| ALLEN  |   61439920.000 | 240.00 | 160.0 |  80.00 |
| TURNER |   50624925.000 | 225.00 | 150.0 |  75.00 |
| MILLER |   32954935.000 | 195.00 | 130.0 |  65.00 |
| MARTIN |   29296812.500 | 187.50 | 125.0 |  62.50 |
| WARD   |   29296812.500 | 187.50 | 125.0 |  62.50 |
| ADAMS  |   19964945.000 | 165.00 | 110.0 |  55.00 |
| JAMES  |   12860577.500 | 142.50 |  95.0 |  47.50 |
| SMITH  |    7679960.000 | 120.00 |  80.0 |  40.00 |
+--------+---------------+--------+-------+--------+
14 rows in set (0.001 sec)
```

8. Update the salary of each employee by 10% increment who are not eligible for commission.

```
MariaDB [ritika0748]> UPDATE employee
    -> SET SAL = SAL * 1.1
    -> WHERE COMM IS NULL OR COMM = 0;
Query OK, 11 rows affected (0.021 sec)
Rows matched: 11  Changed: 11  Warnings: 0

MariaDB [ritika0748]> select * from employee;
+-------+--------+-----------+------+------------+------+------+--------+
| EMPNO | ENAME  | JOB       | MGR  | HIREDATE   | SAL  | COMM | DEPTNO |
+-------+--------+-----------+------+------------+------+------+--------+
|  7369 | SMITH  | CLERK     | 7902 | 1980-12-17 |  968 | NULL |     20 |
|  7499 | ALLEN  | SALESMAN  | 7698 | 1981-02-20 | 1600 |  300 |     30 |
|  7521 | WARD   | SALESMAN  | 7698 | 1981-02-22 | 1250 |  300 |     30 |
|  7566 | JONES  | MANAGER   | 7839 | 1981-04-02 | 3600 | NULL |     20 |
|  7654 | MARTIN | SALESMAN  | 7698 | 1981-09-28 | 1250 | 1400 |     30 |
|  7698 | BLAKE  | MANAGER   | 7839 | 1981-05-01 | 3449 | NULL |     30 |
|  7782 | CLARK  | MANAGER   | 7839 | 1981-06-09 | 2965 | NULL |     20 |
|  7788 | SCOTT  | ANALYST   | 7566 | 1982-12-09 | 3630 | NULL |     40 |
|  7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 6050 | NULL |     20 |
|  7844 | TURNER | SALESMAN  | 7698 | 1981-09-08 | 1815 |    0 |     30 |
|  7876 | ADAMS  | CLERK     | 7788 | 1983-01-12 | 1331 | NULL |     20 |
|  7900 | JAMES  | CLERK     | 7698 | 1981-12-03 | 1150 | NULL |     30 |
|  7902 | FORD   | ANALYST   | 7566 | 1981-12-03 | 3630 | NULL |     20 |
|  7934 | MILLER | CLERK     | 7782 | 1982-01-23 | 1573 | NULL |     10 |
+-------+--------+-----------+------+------------+------+------+--------+
14 rows in set (0.001 sec)
```

9. Display those employees whose salary is more than 3000 after giving 20% increment.

```
MariaDB [ritik0884]> SELECT ename ,sal , sal*1.20 as new_salary
    -> FROM employee where sal*1.20>3000;
+-------+------+------------+
| ename | sal  | new_salary |
+-------+------+------------+
| JONES | 3273 |    3927.60 |
| BLAKE | 3135 |    3762.00 |
| CLARK | 2695 |    3234.00 |
| SCOTT | 3300 |    3960.00 |
| KING  | 5500 |    6600.00 |
| FORD  | 3300 |    3960.00 |
+-------+------+------------+
6 rows in set (0.001 sec)
```

10. Display those employees whose salary contains atleast 3 digits.

```
MariaDB [ritika0748]> select ENAME, SAL from employee
    -> where length(SAL) >= 3;
+--------+------+
| ENAME  | SAL  |
+--------+------+
| SMITH  |  968 |
| ALLEN  | 1600 |
| WARD   | 1250 |
| JONES  | 3600 |
| MARTIN | 1250 |
| BLAKE  | 3449 |
| CLARK  | 2965 |
| SCOTT  | 3630 |
| KING   | 6050 |
| TURNER | 1815 |
| ADAMS  | 1331 |
| JAMES  | 1150 |
| FORD   | 3630 |
| MILLER | 1573 |
+--------+------+
14 rows in set (0.001 sec)
```

EXPERIMENT NO. 05

1. Display the total number of employee working in the company.

```
MariaDB [ritika0748]> select count(*) AS TotalEmployee from employee;
+---------------+
| TotalEmployee |
+---------------+
|            14 |
+---------------+
1 row in set (0.005 sec)
```

2. Display the total salary being paid to all employees.

```
MariaDB [ritika0748]> select sum(sal) AS TotalSAL from employee;
+----------+
| TotalSAL |
+----------+
|    29025 |
+----------+
1 row in set (0.003 sec)
```

3. Display the maximum salary from employee table.

```
MariaDB [ritika0748]> select max(sal) AS MaximumSalary from employee;
+---------------+
| MaximumSalary |
+---------------+
|          5000 |
+---------------+
1 row in set (0.001 sec)
```

4. Display the minimum salary from employee table.

```
MariaDB [ritika0748]> select min(sal) AS MinimumSalary from employee;
+---------------+
| MinimumSalary |
+---------------+
|           800 |
+---------------+
1 row in set (0.001 sec)
```

5. Display the average salary from employee table

```
MariaDB [ritika0748]> select avg(sal) AS AverageSAL from employee;
+------------+
| AverageSAL |
+------------+
|  2073.2143 |
+------------+
1 row in set (0.001 sec)
```

6. Display the maximum salary being paid to clerk.

```
MariaDB [ritika0748]> select max(sal) AS MaximumSalary_CLERK from employee
    -> where JOB = 'Clerk';
+---------------------+
| MaximumSalary_CLERK |
+---------------------+
|                1300 |
+---------------------+
1 row in set (0.001 sec)
```

7. Display the maximum salary being paid in dept no 20.

```
MariaDB [ritika0748]> select max(sal) AS MaximumSalary_inDept20 from employee
    -> where DeptNo = 20;
+------------------------+
| MaximumSalary_inDept20 |
+------------------------+
|                   5000 |
+------------------------+
1 row in set (0.005 sec)
```

8. Display the minimum salary paid to any salesman.

```
MariaDB [ritika0748]> select min(sal) AS MinimumSalary_Salesman from employee
    -> where JOB = 'Salesman';
+------------------------+
| MinimumSalary_Salesman |
+------------------------+
|                   1250 |
+------------------------+
1 row in set (0.001 sec)
```

9. Display the average salary drawn by managers.

```
MariaDB [ritika0748]> select avg(sal) AS AverageSal_Managers from employee
    -> where JOB = 'Manager';
+----------------------+
| AverageSal_Managers |
+----------------------+
|           2758.3333 |
+----------------------+
1 row in set (0.001 sec)
```

10. Display the total salary drawn by analyst working in dept no 40.

```
MariaDB [ritika0748]> select sum(sal) AS TotalSal_Analyst_inDept40 from employee
    -> where JOB = 'Analyst' and DeptNo = 40;
+---------------------------+
| TotalSal_Analyst_inDept40 |
+---------------------------+
|                      3000 |
+---------------------------+
1 row in set (0.001 sec)
```

11. Display the names of the employee in Uppercase.

```
MariaDB [ritika0748]> select upper(ename) from employee;
+--------------+
| upper(ename) |
+--------------+
| SMITH        |
| ALLEN        |
| WARD         |
| JONES        |
| MARTIN       |
| BLAKE        |
| CLARK        |
| SCOTT        |
| KING         |
| TURNER       |
| ADAMS        |
| JAMES        |
| FORD         |
| MILLER       |
+--------------+
14 rows in set (0.001 sec)
```

```
MariaDB [ritika0748]> select ucase(ename) from employee;
+--------------+
| ucase(ename) |
+--------------+
| SMITH        |
| ALLEN        |
| WARD         |
| JONES        |
| MARTIN       |
| BLAKE        |
| CLARK        |
| SCOTT        |
| KING         |
| TURNER       |
| ADAMS        |
| JAMES        |
| FORD         |
| MILLER       |
+--------------+
14 rows in set (0.001 sec)
```

12. Display the names of the employee in Lowercase.

```
MariaDB [ritika0748]> select lower(ename) from employee;
+--------------+
| lower(ename) |
+--------------+
| smith        |
| allen        |
| ward         |
| jones        |
| martin       |
| blake        |
| clark        |
| scott        |
| king         |
| turner       |
| adams        |
| james        |
| ford         |
| miller       |
+--------------+
14 rows in set (0.001 sec)
```

```
MariaDB [ritika0748]> select lcase(ename) from employee;
+--------------+
| lcase(ename) |
+--------------+
| smith        |
| allen        |
| ward         |
| jones        |
| martin       |
| blake        |
| clark        |
| scott        |
| king         |
| turner       |
| adams        |
| james        |
| ford         |
| miller       |
+--------------+
14 rows in set (0.001 sec)
```

13. Display the names of the employee in Proper case.

```
MariaDB [ritika0748]> select concat(substring(ename, 1,1), lower(substring(ename, 2)))
    -> AS ProperName from employee;
+------------+
| ProperName |
+------------+
| Smith      |
| Allen      |
| Ward       |
| Jones      |
| Martin     |
| Blake      |
| Clark      |
| Scott      |
| King       |
| Turner     |
| Adams      |
| James      |
| Ford       |
| Miller     |
+------------+
14 rows in set (0.001 sec)
```

14. Display the length of Your name using appropriate function.

```
MariaDB [ritika0748]> select LENGTH('RITIKA') AS my_name_length;
+----------------+
| my_name_length |
+----------------+
|              6 |
+----------------+
1 row in set (0.018 sec)
```

15. Display the length of all the employee names.

```
MariaDB [ritika0748]> select ename, Length(ename) AS name_length from employee;
+--------+-------------+
| ename  | name_length |
+--------+-------------+
| SMITH  |           5 |
| ALLEN  |           5 |
| WARD   |           4 |
| JONES  |           5 |
| MARTIN |           6 |
| BLAKE  |           5 |
| CLARK  |           5 |
| SCOTT  |           5 |
| KING   |           4 |
| TURNER |           6 |
| ADAMS  |           5 |
| JAMES  |           5 |
| FORD   |           4 |
| MILLER |           6 |
+--------+-------------+
14 rows in set (0.003 sec)
```

EXPERIMENT NO. 06

1. Display empno, ename, deptno from employee table. Instead of display department numbers display the related department name (Use decode function).

```
MariaDB [ritika0748]> select ename, empno, deptno, case deptno
    -> when 10 then 'research'
    -> when 20 then 'accounting'
    -> when 30 then 'sales'
    -> when 40 then 'operations'
    -> else 'Unknown Department'
    -> end as departmentName
    -> from employee;
+--------+-------+--------+----------------+
| ename  | empno | deptno | departmentName |
+--------+-------+--------+----------------+
| SMITH  |  7369 |     20 | accounting     |
| ALLEN  |  7499 |     30 | sales          |
| WARD   |  7521 |     30 | sales          |
| JONES  |  7566 |     20 | accounting     |
| MARTIN |  7654 |     30 | sales          |
| BLAKE  |  7698 |     30 | sales          |
| CLARK  |  7782 |     20 | accounting     |
| SCOTT  |  7788 |     40 | operations     |
| KING   |  7839 |     20 | accounting     |
| TURNER |  7844 |     30 | sales          |
| ADAMS  |  7876 |     20 | accounting     |
| JAMES  |  7900 |     30 | sales          |
| FORD   |  7902 |     20 | accounting     |
| MILLER |  7934 |     10 | research       |
+--------+-------+--------+----------------+
14 rows in set (0.023 sec)
```

2. Display your age in days.

```
MariaDB [ritika0748]> select datediff ('2026-02-24','2006-07-09') AS MyAgeInDays;
+-------------+
| MyAgeInDays |
+-------------+
|        7170 |
+-------------+
1 row in set (0.018 sec)

MariaDB [ritika0748]> select datediff (curdate(),'2006-07-09') AS MyAgeInDays;
+-------------+
| MyAgeInDays |
+-------------+
|        7170 |
+-------------+
1 row in set (0.000 sec)
```

3. Display your age in months. 1,2

```
MariaDB [ritika0748]> select period_diff(date_format('2026-02-24','%Y%m'),
    -> date_format('2006-07-09','%Y%m')) AS MyAgeInMonths;
+---------------+
| MyAgeInMonths |
+---------------+
|           235 |
+---------------+
1 row in set (0.001 sec)

MariaDB [ritika0748]> select period_diff(date_format(curdate(),'%Y%m'),
    -> date_format('2006-07-09','%Y%m')) AS MyAgeInMonths;
+---------------+
| MyAgeInMonths |
+---------------+
|           235 |
+---------------+
1 row in set (0.000 sec)
```

4. Display the current date as 15th August Friday Nineteen Ninety-Seven.

```
MariaDB [ritika0748]> select CONCAT(date_format(curdate(),"%D %M %W "),"Twenty Twenty-Six")
AS FormattedDate;
+----------------------------------------+
| FormattedDate                          |
+----------------------------------------+
| 24th February Tuesday Twenty Twenty-Six |
+----------------------------------------+
1 row in set (0.000 sec)
```

5. Display the following output for each row from employee table.
6. Scott has joined the company on Wednesday 13th August Nineteen Ninety

```
MariaDB [ritika0748]> select ename, concat(ename, " has joined the company on ", date_format
(hiredate, "%W %D %M %Y")) as FORMATTED_INFO from EMPLOYEE;
+--------+-----------------------------------------------------------------+
| ename  | FORMATTED_INFO                                                  |
+--------+-----------------------------------------------------------------+
| SMITH  | SMITH has joined the company on Wednesday 17th December 1980 |
| ALLEN  | ALLEN has joined the company on Friday 20th February 1981    |
| WARD   | WARD has joined the company on Sunday 22nd February 1981     |
| JONES  | JONES has joined the company on Thursday 2nd April 1981      |
| MARTIN | MARTIN has joined the company on Monday 28th September 1981  |
| BLAKE  | BLAKE has joined the company on Friday 1st May 1981          |
| CLARK  | CLARK has joined the company on Tuesday 9th June 1981        |
| SCOTT  | SCOTT has joined the company on Thursday 9th December 1982   |
| KING   | KING has joined the company on Tuesday 17th November 1981    |
| TURNER | TURNER has joined the company on Tuesday 8th September 1981  |
| ADAMS  | ADAMS has joined the company on Wednesday 12th January 1983  |
| JAMES  | JAMES has joined the company on Thursday 3rd December 1981   |
| FORD   | FORD has joined the company on Thursday 3rd December 1981    |
| MILLER | MILLER has joined the company on Saturday 23rd January 1982  |
+--------+-----------------------------------------------------------------+
14 rows in set (0.001 sec)
```

7. Find the date for nearest Saturday after current date.

```
MariaDB [ritika0748]> SELECT CASE
    -> WHEN WEEKDAY(CURDATE()) <= 5 THEN DATE_ADD(CURDATE(), INTERVAL (5-WEEKDAY(CURDATE())) DAY)
    -> ELSE
    -> DATE_ADD(CURDATE(), INTERVAL(12 - WEEKDAY(CURDATE())) DAY)
    -> END
    -> AS NEXT_SATURDAY;
+---------------+
| NEXT_SATURDAY |
+---------------+
| 2026-02-28    |
+---------------+
1 row in set (0.001 sec)
```

8. Display current time.

```
MariaDB [ritika0748]> SELECT CURTIME();
+-----------+
| CURTIME() |
+-----------+
| 00:14:25  |
+-----------+
1 row in set (0.017 sec)
```

9. Display the date three months Before the current date

```
MariaDB [ritika0748]> SELECT DATE_SUB(CURDATE(), INTERVAL 3 MONTH) AS PASTDATE;
+------------+
| PASTDATE   |
+------------+
| 2025-11-25 |
+------------+
1 row in set (0.000 sec)
```

10. Display those employees who joined in the company in the month of Dec.

```
MariaDB [ritika0748]> SELECT ENAME, HIREDATE FROM EMPLOYEE
    -> WHERE MONTH (HIREDATE) = 12;
+-------+------------+
| ENAME | HIREDATE   |
+-------+------------+
| SMITH | 1980-12-17 |
| SCOTT | 1982-12-09 |
| JAMES | 1981-12-03 |
| FORD  | 1981-12-03 |
+-------+------------+
4 rows in set (0.018 sec)
```

11. Display those employees whose first 2 characters from hire date -last 2 characters of salary.

```
MariaDB [ritika0748]> SELECT ENAME, HIREDATE, SAL FROM EMPLOYEE
    -> WHERE LEFT(HIREDATE,2) = RIGHT(SAL,2);
Empty set (0.017 sec)
```

12. Display those employees whose 10% of salary is equal to the year of joining.

```
MariaDB [ritika0748]> SELECT ENAME, HIREDATE, SAL FROM EMPLOYEE
    -> WHERE YEAR (HIREDATE) = SAL * 0.1;
Empty set (0.000 sec)
```

13. Display those employees who joined the company before 15 of the months.

```
MariaDB [ritika0748]> SELECT ENAME, HIREDATE FROM EMPLOYEE
    -> WHERE DAY(HIREDATE) < 15;
+--------+------------+
| ENAME  | HIREDATE   |
+--------+------------+
| JONES  | 1981-04-02 |
| BLAKE  | 1981-05-01 |
| CLARK  | 1981-06-09 |
| SCOTT  | 1982-12-09 |
| TURNER | 1981-09-08 |
| ADAMS  | 1983-01-12 |
| JAMES  | 1981-12-03 |
| FORD   | 1981-12-03 |
+--------+------------+
8 rows in set (0.019 sec)
```

14. Display those employees who has joined before 15th of the month


```
MariaDB [ritika0748]> SELECT ENAME, HIREDATE FROM EMPLOYEE
    -> WHERE DAY(HIREDATE) < 15;
+--------+------------+
| ENAME  | HIREDATE   |
+--------+------------+
| JONES  | 1981-04-02 |
| BLAKE  | 1981-05-01 |
| CLARK  | 1981-06-09 |
| SCOTT  | 1982-12-09 |
| TURNER | 1981-09-08 |
| ADAMS  | 1983-01-12 |
| JAMES  | 1981-12-03 |
| FORD   | 1981-12-03 |
+--------+------------+
8 rows in set (0.019 sec)
```


15. Display those employees whose joining DATE is available in deptno.

```
MariaDB [ritika0748]> SELECT ENAME, DEPTNO, HIREDATE FROM EMPLOYEE
    -> WHERE DAY(HIREDATE) IN (10,20,30,40);
+-------+--------+------------+
| ENAME | DEPTNO | HIREDATE   |
+-------+--------+------------+
| ALLEN |     30 | 1981-02-20 |
+-------+--------+------------+
1 row in set (0.017 sec)
```

EXPERIMENT NO. 07

1. Compute the no. of days remaining in this year.

```
MariaDB [ritika0748]> select datediff('2026-12-31', curdate()) AS RemainingDays;
+---------------+
| RemainingDays |
+---------------+
|           308 |
+---------------+
1 row in set (0.001 sec)
```

2. Find the highest and lowest salaries and the difference between of them.

```
MariaDB [ritika0748]> select max(sal), min(sal), max(sal) - min(sal) AS Difference from employee;
+----------+----------+------------+
| max(sal) | min(sal) | Difference |
+----------+----------+------------+
|     5000 |      800 |       4200 |
+----------+----------+------------+
1 row in set (0.001 sec)
```

3. List employee whose commission is greater than 25 % of their salaries.

```
MariaDB [ritika0748]> select ename, sal, comm from employee
    -> where comm > sal * 0.25;
+--------+------+------+
| ename  | sal  | comm |
+--------+------+------+
| MARTIN | 1250 | 1400 |
+--------+------+------+
1 row in set (0.001 sec)
```

4. Make a query that displays salary in dollar format.

```
MariaDB [ritika0748]> select ename, concat('$',format(SAL,2)) AS SALARY_DOLLARS from employee;
+--------+----------------+
| ename  | SALARY_DOLLARS |
+--------+----------------+
| SMITH  | $800.00        |
| ALLEN  | $1,600.00      |
| WARD   | $1,250.00      |
| JONES  | $2,975.00      |
| MARTIN | $1,250.00      |
| BLAKE  | $2,850.00      |
| CLARK  | $2,450.00      |
| SCOTT  | $3,000.00      |
| KING   | $5,000.00      |
| TURNER | $1,500.00      |
| ADAMS  | $1,100.00      |
| JAMES  | $950.00        |
| FORD   | $3,000.00      |
| MILLER | $1,300.00      |
+--------+----------------+
14 rows in set (0.001 sec)
```

5. Create a matrix query to display the job, the salary for that job based on department number, and the total salary for that job for all departments, giving each column an appropriate heading.

```
MariaDB [ritika0748]> SELECT JOB,
    -> SUM(CASE WHEN DEPTNO = 10 THEN SAL ELSE 0 END) AS D10_TotalSal,
    -> SUM(CASE WHEN DEPTNO = 20 THEN SAL ELSE 0 END) AS D20_TotalSal,
    -> SUM(CASE WHEN DEPTNO = 30 THEN SAL ELSE 0 END) AS D30_TotalSal,
    -> SUM(CASE WHEN DEPTNO = 40 THEN SAL ELSE 0 END) AS D40_TotalSal,
    -> SUM(SAL) AS TOTAL_SAL
    -> FROM EMPLOYEE
    -> GROUP BY JOB;
+-----------+--------------+--------------+--------------+--------------+-----------+
| JOB       | D10_TotalSal | D20_TotalSal | D30_TotalSal | D40_TotalSal | TOTAL_SAL |
+-----------+--------------+--------------+--------------+--------------+-----------+
| ANALYST   |            0 |         3000 |            0 |         3000 |      6000 |
| CLERK     |         1300 |         1900 |          950 |            0 |      4150 |
| MANAGER   |            0 |         5425 |         2850 |            0 |      8275 |
| PRESIDENT |            0 |         5000 |            0 |            0 |      5000 |
| SALESMAN  |            0 |            0 |         5600 |            0 |      5600 |
+-----------+--------------+--------------+--------------+--------------+-----------+
5 rows in set (0.001 sec)
```

6. Query that will display the total no of employees, and of that total the number who were hired in 1980,1981,1982 and 1983. Give appropriate column heading.

```
MariaDB [ritika0748]> SELECT COUNT(*) AS TOTAL_EMP,
    -> SUM(CASE WHEN YEAR(HIREDATE) = 1980 THEN 1 ELSE 0 END) AS 1980_END,
    -> SUM(CASE WHEN YEAR(HIREDATE) = 1980 THEN 1 ELSE 0 END) AS 1981_END,
    -> SUM(CASE WHEN YEAR(HIREDATE) = 1980 THEN 1 ELSE 0 END) AS 1982_END,
    -> SUM(CASE WHEN YEAR(HIREDATE) = 1980 THEN 1 ELSE 0 END) AS 1983_END
    -> FROM EMPLOYEE;
+-----------+----------+----------+----------+----------+
| TOTAL_EMP | 1980_END | 1981_END | 1982_END | 1983_END |
+-----------+----------+----------+----------+----------+
|        14 |        1 |        1 |        1 |        1 |
+-----------+----------+----------+----------+----------+
1 row in set (0.026 sec)
```

7. Query to get the last Sunday of Any Month.

```
MariaDB [ritika0748]> SELECT LAST_DAY('2026-02-26') - INTERVAL (WEEKDAY(LAST_DAY('2026-02-26'))+1) % 7 DAY AS LAST_SUNDAY;
+-------------+
| LAST_SUNDAY |
+-------------+
| 2026-02-22  |
+-------------+
1 row in set (0.001 sec)
```

8. Display department numbers and total number of employees working in each department.

```
MariaDB [ritika0748]> select DEPTNO, count(DEPTNO) AS TotalEmployees from employee
    -> group by DEPTNO;
+--------+----------------+
| DEPTNO | TotalEmployees |
+--------+----------------+
|     10 |              1 |
|     20 |              6 |
|     30 |              6 |
|     40 |              1 |
+--------+----------------+
4 rows in set (0.001 sec)
```

9. Display the various jobs and total number of employees within each job group.

```
MariaDB [ritika0748]> select JOB, count(JOB) AS TotalEmployees from employee
    -> group by JOB;
+-----------+----------------+
| JOB       | TotalEmployees |
+-----------+----------------+
| ANALYST   |              2 |
| CLERK     |              4 |
| MANAGER   |              3 |
| PRESIDENT |              1 |
| SALESMAN  |              4 |
+-----------+----------------+
5 rows in set (0.002 sec)
```

10. Display the depart numbers and total salary for each department.

```
MariaDB [ritika0748]> select DEPTNO, sum(sal) AS TotalSALARY from employee
    -> group by DEPTNO;
+--------+-------------+
| DEPTNO | TotalSALARY |
+--------+-------------+
|     10 |        1300 |
|     20 |       15325 |
|     30 |        9400 |
|     40 |        3000 |
+--------+-------------+
4 rows in set (0.001 sec)
```